

Bathymetric Terrain Analysis of the Catalan Coast

Data Exploration Report

SotaMar — Spatial Data Infrastructure for Dive Site Cataloging

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Abstract

This report documents the exploration of high-resolution (1 m) bathymetric data from the Institut Cartogràfic i Geològic de Catalunya (ICGC) for dive site terrain characterization along the Catalan coast. We describe the data access methodology via Cloud Optimized GeoTIFF (COG) remote reading, perform a detailed pilot analysis at the Illes Medes Marine Protected Area, and extend the terrain analysis to six sites spanning the full 240 km coastline. Four morphometrics are computed at each site: slope, fine-scale and broad-scale Bathymetric Position Index (BPI), and Vector Ruggedness Measure (VRM). Cross-site comparison reveals that VRM provides a consistent gradient of terrain complexity across all six sites, aligning well with known dive site character. A case study of El Biotop artificial reef (Torredembarra) exposes a temporal limitation: the 22 m-tall structure is absent from the data, indicating that survey acquisition predates its 2023 construction.

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1 Introduction

The SotaMar project aims to build a spatial data infrastructure (SDI) for cataloging recreational scuba dive sites along the Catalan coast, integrating bathymetric terrain analysis with dive site metadata. Before designing the system architecture, we perform a data exploration phase to understand the available bathymetric data and validate that derived terrain metrics can capture dive-relevant geomorphological features.

The Illes Medes (Medes Islands) near L’Estartit were chosen as the pilot area due to their status as a Marine Protected Area, their complex underwater topography (walls, tunnels, pinacles), and their importance as a dive tourism destination with approximately 80,000 dives per year [1].

2 Data Source: ICGC Bathymetry

2.1 Dataset Description

The ICGC distributes a coastal bathymetric-topographic elevation model covering the Catalan littoral fringe as a single Cloud Optimized GeoTIFF (COG). Table 1 summarizes the dataset characteristics obtained by remote inspection.

Table 1: ICGC bathymetry COG metadata.

Property	Value
Format	GeoTIFF (Cloud Optimized)
File size	3.3 GB
Grid dimensions	238 202 × 223 252 pixels
Resolution	1 m × 1 m
Data type	32-bit float
CRS	EPSG:25831 (ETRS89 / UTM zone 31N)
Compression	Deflate
Overview levels	9 (2× to 512×)
NoData value	−9999
Vertical datum	EGM08D595 geoid (orthometric heights)
Full extent (E)	289 416 m to 527 618 m
Full extent (N)	4 475 476 m to 4 698 728 m

2.2 COG URL Discovery

The dataset URL was not documented in the ICGC’s WMS GetCapabilities response or public metadata catalogs. It was discovered by browsing the Apache directory index at `datacloud.icgc.cat/datacloud` which revealed the file:

`batimetria-v2r1-elevacions-2021-2025.tif`

The inclusion of `elevacions` in the filename—absent from other ICGC product naming patterns—was the key discovery obstacle. An automated probing script was written to systematically try candidate URL patterns before falling back to directory listing inspection.

2.3 Remote COG Access

The COG format supports HTTP range requests, allowing `rasterio` (via GDAL’s `/vsicurl/` virtual filesystem) to read arbitrary spatial subsets without downloading the full file. The following GDAL configuration was used for optimal performance:

```

1 from rasterio.env import Env
2 from rasterio.windows import from_bounds
3
4 with Env(GDAL_DISABLE_READDIR_ON_OPEN="EMPTY_DIR",
5         GDAL_HTTP_MERGE_CONSECUTIVE_RANGES="YES",
6         VSI_CACHE=True, VSI_CACHE_SIZE=5_000_000):
7     with rasterio.open(COG_URL) as src:
8         window = from_bounds(517400, 4654100, 519400, 4656100,
9                             transform=src.transform)
10        data = src.read(1, window=window)

```

This extracted a 2000×2000 pixel window in under 10 seconds over a standard network connection. The extracted subset was saved as a local GeoTIFF with LZW compression (8.9 MB) for subsequent analysis.

3 Study Area: Illes Medes

The extraction window covers a $2\text{ km} \times 2\text{ km}$ area centered on the Medes Islands (approximate center: 518 400 E, 4655 100 N in EPSG:25831). Table 2 summarizes the elevation statistics.

Table 2: Elevation statistics for the Illes Medes extraction window.

Statistic	Value
Minimum	−61.55 m
Maximum	86.34 m
Mean	−23.69 m
Standard dev.	23.41 m
NoData pixels	724 433 (18.1%)

The maximum elevation of 86.3 m corresponds to the summit of Meda Gran, the largest island in the archipelago, confirming that the ICGC dataset includes topography above the geoid, not only submerged bathymetry. The strongly negative mean (−23.7 m) reflects the predominantly submarine character of the window, with the island group occupying only a fraction of the 4 km^2 area. The 18.1% NoData corresponds to the deep-water western margin beyond the survey’s littoral fringe.

Figure 1 shows the extracted bathymetry with 5 m contour lines and a corresponding hillshade visualization.

4 Terrain Morphometrics

Four terrain metrics were computed from the elevation grid, all at native 1 m resolution. NoData areas were handled by masking prior to computation and propagating NoData to output cells where valid neighbors were insufficient.

4.1 Slope

Slope magnitude was computed from the first-order partial derivatives of the elevation surface using central finite differences [4]:

$$\text{slope} = \arctan \left(\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \right) \quad (1)$$

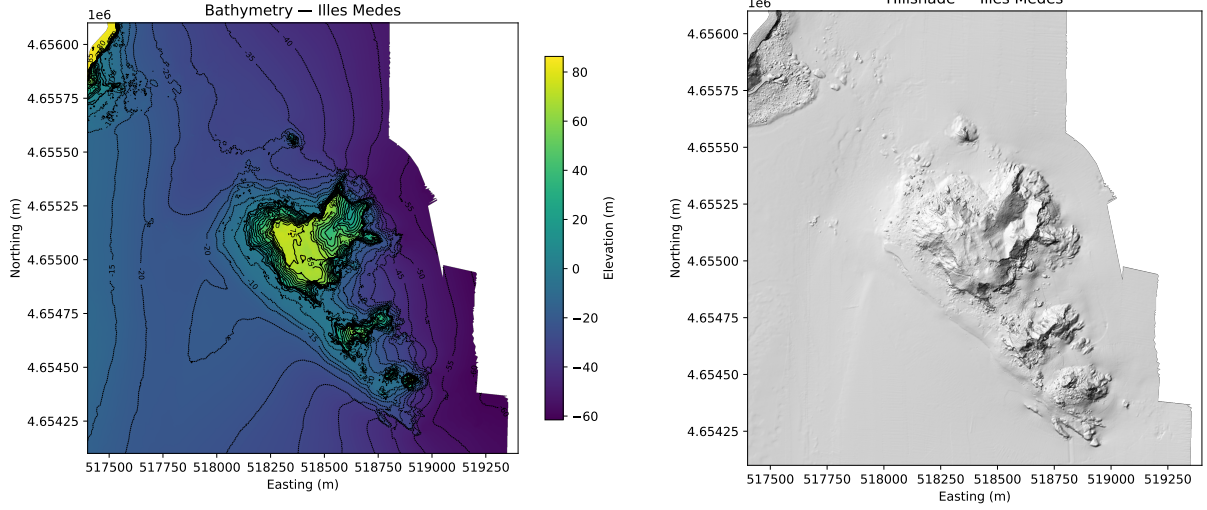


Figure 1: Illes Medes bathymetry. **Left:** Elevation with 5 m contour intervals (viridis colormap). **Right:** Analytical hillshade (azimuth 315° , altitude 45° , vertical exaggeration $2\times$). The Medes archipelago is clearly visible as emerged outcrops in the center-east, with steep submarine walls descending to -50 m and beyond. The complex island topography—walls, pinnacles, and inter-island channels—is well resolved at 1 m resolution.

With the 1 m grid spacing, $\partial z/\partial x$ and $\partial z/\partial y$ were obtained via `numpy.gradient`:

```
1 dz_dy, dz_dx = np.gradient(elevation, 1.0, 1.0)
2 slope_rad = np.arctan(np.sqrt(dz_dx**2 + dz_dy**2))
3 slope_deg = np.degrees(slope_rad)
```

The resulting slope ranges from 0° to 88.9° (mean 6.6° , $\sigma = 12.3^\circ$). The near-vertical maxima occur at the cliff faces of the islands, which represent classic wall dive topography.

4.2 Bathymetric Position Index (BPI)

The Bathymetric Position Index [2] quantifies whether a cell sits on a local ridge (positive), in a depression (negative), or on a flat/slope (near zero):

$$\text{BPI} = z_{\text{cell}} - \bar{z}_{\text{annulus}} \quad (2)$$

where $\bar{z}_{\text{annulus}}$ is the mean elevation within an annular neighborhood of inner radius r_i and outer radius r_o . Two scales were computed:

- **Fine scale** ($r_i = 3$ m, $r_o = 5$ m): captures meter-scale features such as individual boulders and small ridges.
- **Broad scale** ($r_i = 25$ m, $r_o = 50$ m): captures the large-scale terrain position— island platforms, inter-island channels, shelf breaks.

The annular mean was computed via 2D convolution with an explicit annular kernel, using a separate validity mask convolution to correct for NoData at data boundaries:

```
1 # Boolean annular kernel
2 y, x = np.ogrid[-r_o:r_o+1, -r_o:r_o+1]
3 annulus = ((np.sqrt(x**2 + y**2) >= r_i)
4             & (np.sqrt(x**2 + y**2) <= r_o)).astype(float)
```

```

5
6 # NoData-aware convolution
7 elev_sum = ndimage.convolve(elev_filled, annulus, mode="constant")
8 valid_count = ndimage.convolve(valid_mask, annulus, mode="constant")
9 annular_mean = elev_sum / np.maximum(valid_count, 1.0)
10 bpi = elevation - annular_mean

```

Table 3 summarizes the BPI statistics. The near-zero means confirm correct implementation (BPI is a zero-mean statistic by construction over large areas). The substantially larger spread at broad scale ($\sigma = 4.7$ m vs. 0.7 m) indicates that the dominant terrain structure at Illes Medes operates at the 25–50 m scale, consistent with the geomorphology of rocky island groups.

Table 3: BPI statistics at fine and broad scales.

Scale	Min (m)	Max (m)	Mean (m)	Std (m)
Fine ($r = 3\text{--}5$ m)	−45.949	41.941	0.000	0.739
Broad ($r = 25\text{--}50$ m)	−43.913	81.010	0.029	4.673

4.3 Vector Ruggedness Measure (VRM)

The Vector Ruggedness Measure [3] captures terrain heterogeneity independently of slope magnitude. Each cell’s surface orientation is expressed as a 3D unit normal vector, and VRM measures the dispersion of these vectors within a moving window:

$$\mathbf{n}_i = \begin{pmatrix} \sin(\text{slope}_i) \sin(\text{aspect}_i) \\ \sin(\text{slope}_i) \cos(\text{aspect}_i) \\ \cos(\text{slope}_i) \end{pmatrix}, \quad \text{VRM} = 1 - \frac{|\sum_{i=1}^n \mathbf{n}_i|}{n} \quad (3)$$

A 3×3 window ($n = 9$) was used. The component sums were computed efficiently using `scipy.ndimage.uniform_filter` (a separable $O(N)$ filter) multiplied by n to recover the sum from the mean:

```

1 n = window_size ** 2
2 x_sum = ndimage.uniform_filter(x, size=window_size,
3                               mode="constant") * n
4 y_sum = ndimage.uniform_filter(y, size=window_size,
5                               mode="constant") * n
6 z_sum = ndimage.uniform_filter(z, size=window_size,
7                               mode="constant") * n
8 resultant = np.sqrt(x_sum**2 + y_sum**2 + z_sum**2)
9 vrm = 1.0 - (resultant / n)

```

VRM ranges from 0 (perfectly flat) to 1 (maximally rugged). The observed range of 0 to 0.818 (mean 0.003, $\sigma = 0.022$) indicates that the seafloor is overwhelmingly smooth, with ruggedness concentrated at the island cliff edges. Values above 0.3 cluster exclusively around the steep rocky perimeters—precisely the features that define high-quality wall and cave dive sites.

4.4 Multi-Panel Comparison

Figure 2 presents all five metrics at identical spatial extent for comparison. The complementarity of the metrics is evident: slope highlights gradients, BPI identifies relative position, and VRM isolates terrain heterogeneity.

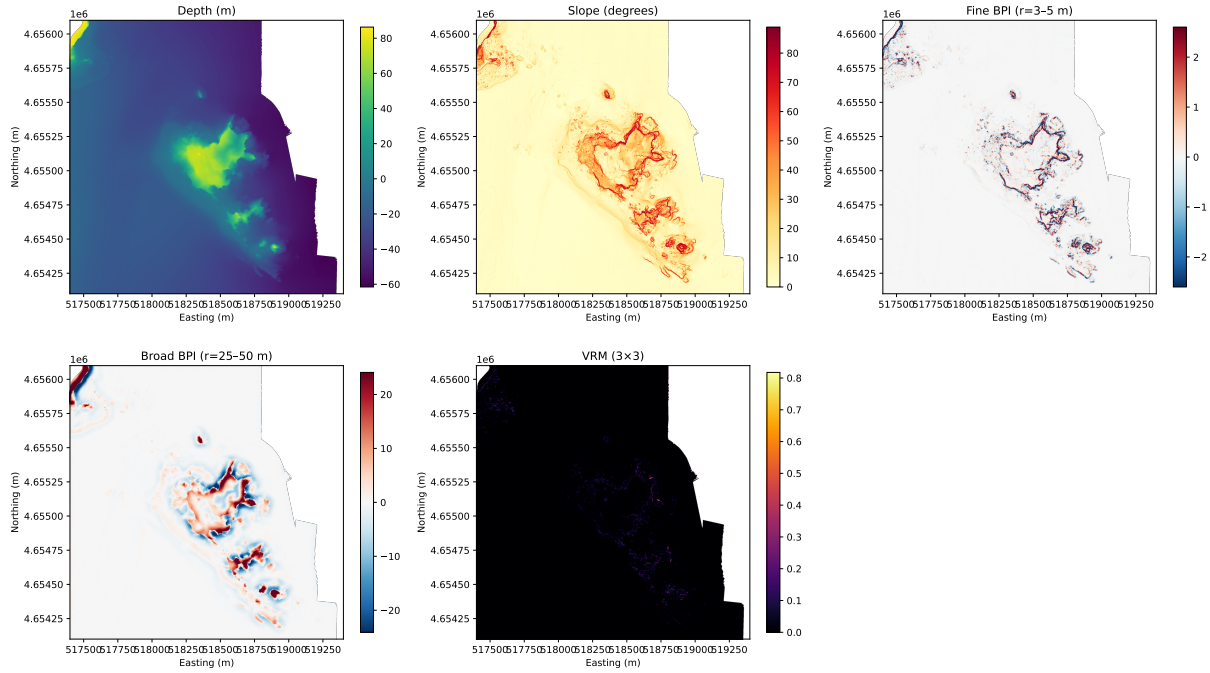


Figure 2: Bathymetric terrain analysis for Illes Medes. **Top row, left to right:** depth (m), slope (degrees), fine-scale BPI ($r = 3\text{--}5\text{ m}$). **Bottom row:** broad-scale BPI ($r = 25\text{--}50\text{ m}$), VRM (3×3 window). BPI panels use a diverging colormap centered at zero (red = ridges, blue = depressions).

5 Depth Profile Analysis

An east–west transect at northing 4655100 m was extracted across the study area, cutting through the heart of the Medes archipelago (Figure 3). The profile reveals:

1. Open water at approximately -20 m depth on the western approach, deepening gradually from the shelf.
2. The emerged island platform rising dramatically from the seafloor to over 70 m above sea level (Meda Gran summit), spanning approximately 400 m horizontally.
3. A steep eastern submarine wall descending from the island back to -50 m over less than 200 m of horizontal distance—a gradient exceeding 30° , characteristic of the dramatic wall dives for which the Medes are renowned.

Recreational diving certification thresholds are annotated: Open Water Diver (OWD, 12 m), Advanced Open Water Diver (AOWD, 18 m), Deep Diver (30 m), and the recreational limit (40 m). The Medes site reaches Deep Diver depths on both flanks and approaches the recreational limit on the east side, confirming the site’s suitability for a range of certification levels from intermediate to advanced.

6 Coast-Wide Analysis

To assess the generalizability of the terrain metrics beyond the Illes Medes pilot area, the analysis was extended to six sites spanning the full length of the Catalan coast (approximately 240 km). Sites were selected to represent a range of coastal geomorphologies: rocky capes, island groups, and gently sloping continental shelf (Figure 4).

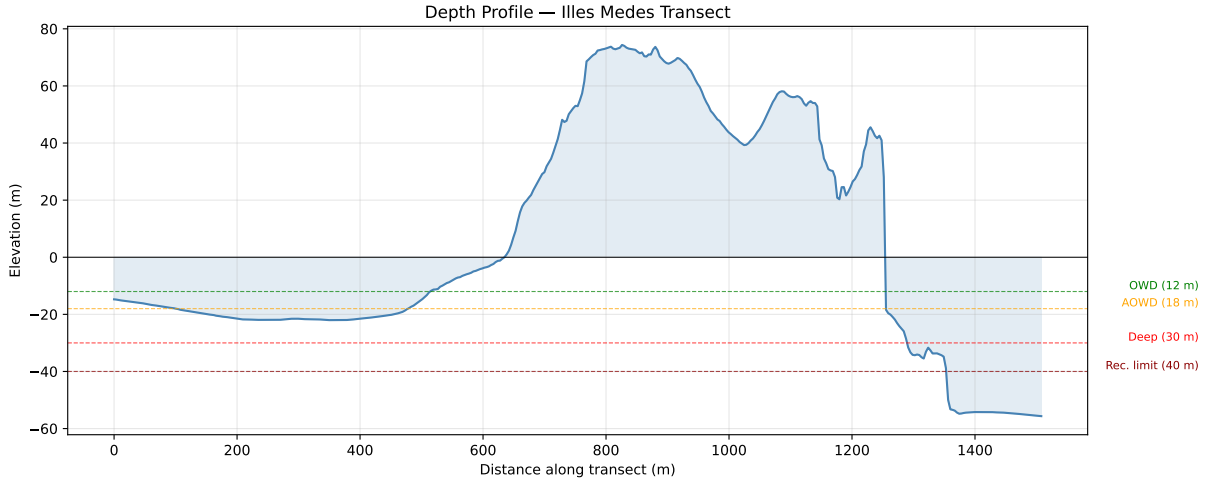


Figure 3: Depth profile along an E–W transect at $N = 4655100$ m, crossing the Medes archipelago. Dashed horizontal lines indicate recreational diving certification depth limits. The dramatic eastern wall (0 m to -50 m over ~ 200 m horizontal) is the defining feature of this internationally renowned dive site.

6.1 Site Selection

Table 4 describes the six study areas. Each site was extracted as a $2\text{ km} \times 2\text{ km}$ window at full 1 m resolution from the remote COG, and all four terrain metrics were computed identically.

Table 4: Study sites along the Catalan coast. Coordinates are window centers in EPSG:25831.

Site	Label	Easting	Northing	Character
1	Roses	520 100	4 677 000	Rocky cape, steep cliffs
2	Illes Medes	518 400	4 655 100	Island MPA, walls & tunnels
3	Illes Formigues	515 400	4 634 500	Rocky islets, gentle shelf
4	Tossa de Mar	495 100	4 618 900	Rocky coast, steep cliffs
5	Costa del Garraf	403 200	4 565 300	Rocky coast, moderate cliffs
6	Cap de Salou	344 800	4 547 200	Rocky cape, moderate relief

6.2 Terrain Character Comparison

Figure 5 shows the hillshade with depth overlay for each site, revealing markedly different terrain character. Roses displays the steepest coastal cliffs, while all sites show a mixture of emerged rocky coast and submerged terrain with varying degrees of complexity.

6.3 Comparative Statistics

Table 5 summarizes the terrain metrics across all sites.

Several patterns emerge:

1. **Roses** is the steepest site, with a mean slope of 12.0° and the highest VRM (0.005). Its combination of steep cliffs and high ruggedness is consistent with its reputation for dramatic cliff-face diving.
2. **All six sites show near-vertical maximum slopes** ($82\text{--}89^\circ$), indicating that even sites with moderate *mean* slope contain localized steep features—rocky outcrops, cliff edges, or

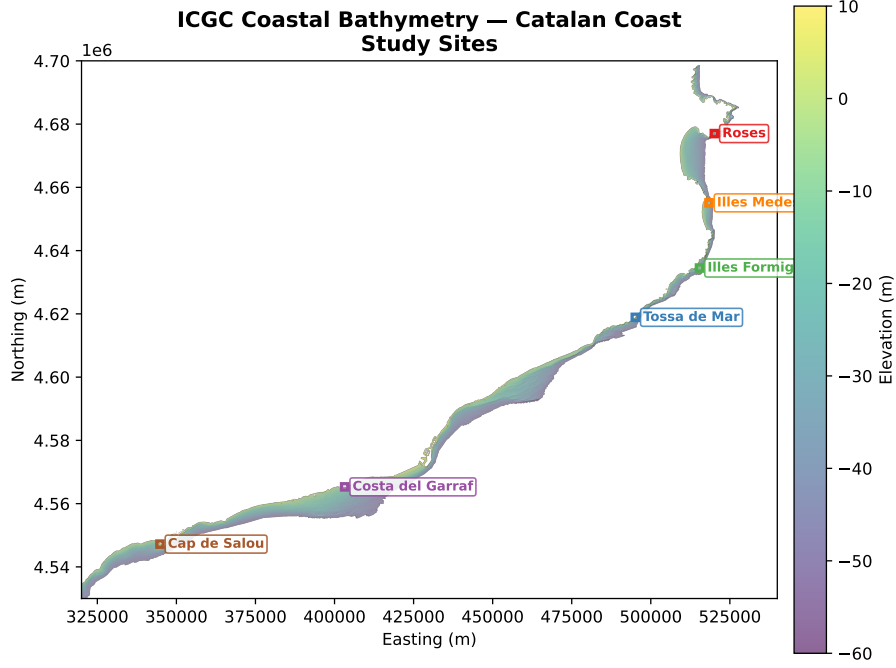


Figure 4: ICGC coastal bathymetry overview (64 m resolution) with the six study sites marked. The dataset covers a narrow littoral fringe from the French border (NE) to Cap de Salou (SW). Background: hillshade with semi-transparent bathymetry overlay.

Table 5: Comparative terrain metrics across the six study sites.

Site	Mean slope ($^{\circ}$)	Max slope ($^{\circ}$)	Mean VRM	Broad BPI σ (m)
Roses	12.0	88.0	0.0050	4.56
Illes Medes	6.6	88.9	0.0030	4.67
Illes Formigues	5.9	82.3	0.0030	1.60
Tossa de Mar	6.6	88.0	0.0040	3.75
Costa del Garraf	3.0	85.2	0.0020	2.45
Cap de Salou	2.7	86.5	0.0030	1.91

coastal bluffs. This suggests that max slope alone is insufficient for site discrimination; mean slope and VRM better capture the overall terrain character.

3. **Broad BPI standard deviation** separates sites more cleanly than mean slope: Illes Medes has the highest σ (4.67 m) alongside Roses (4.56 m), despite lower mean slope, because the islands create strong positive anomalies surrounded by channels. This confirms that BPI captures terrain *position* (ridges vs. depressions) independently of gradient magnitude.
4. **VRM provides a gradient of terrain complexity** across the coast. Roses (0.005) and Tossa de Mar (0.004) lead, followed by the island sites (Medes, Formigues, Salou at 0.003), with Garraf lowest (0.002). Rather than a binary “interesting/featureless” split, the metrics reveal a continuum of terrain complexity that could inform dive site grading systems.

Figure 6 shows the full metric distributions as box plots, confirming the statistical separation.

Terrain Character — Six Dive Areas, Catalan Coast

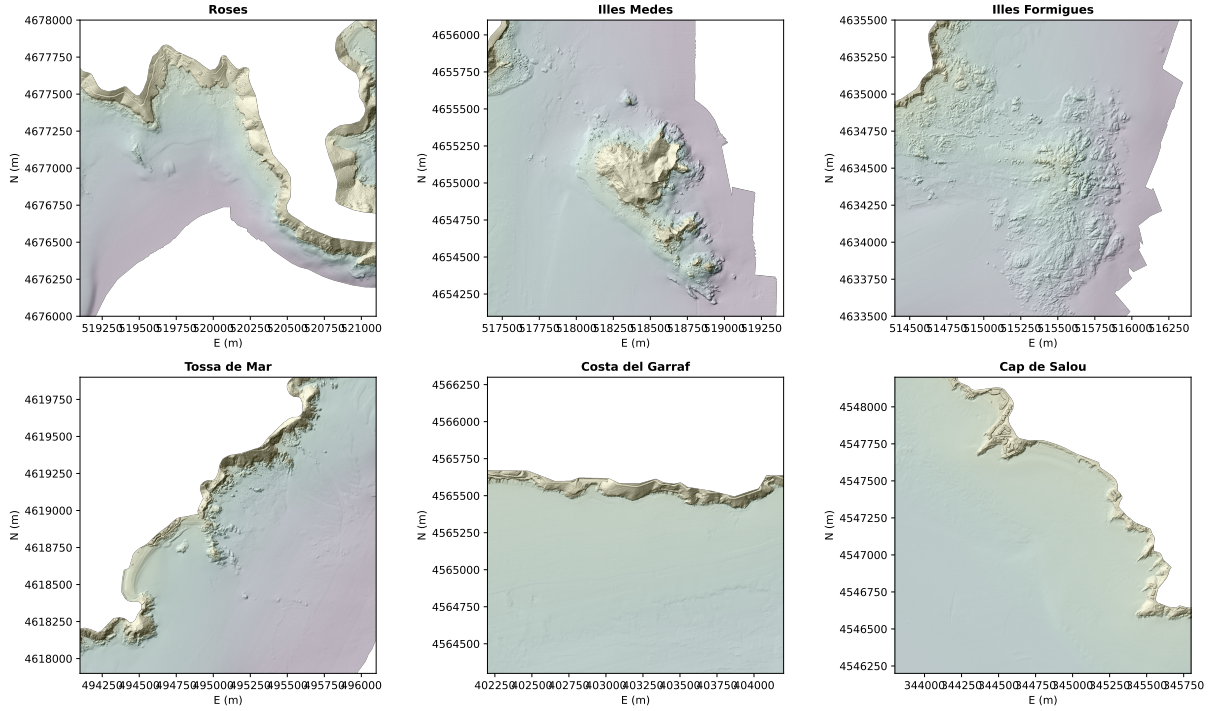


Figure 5: Terrain character at six dive sites along the Catalan coast. Hillshade with semi-transparent bathymetry overlay (viridis). Note the contrast between the steep, rugged terrain at Roses and Tossa de Mar and the smooth shelf at Costa del Garraf.

7 Case Study: El Biotop Artificial Reef

To test the dataset’s ability to detect anthropogenic underwater structures, we examined the location of El Biotop, an artificial reef near Torredembarra (Tarragona). El Biotop is a conical pyramid constructed from approximately 40 000 tonnes of calcium carbonate rock, with its base at -34 m and its peak at -12 m depth—a 22 m vertical structure. Completed in 2023, it has become the only known cleaning station for ocean sunfish (*Mola mola*) in Catalonia, making it a significant dive destination.

7.1 Analysis

The reported coordinates (41.130° N , 1.432° E ; UTM: 368 413 E, 4 554 332 N) were converted to EPSG:25831 and a $500\text{ m} \times 500\text{ m}$ window was extracted at full 1 m resolution (Figure 7). All four terrain metrics were computed.

The results show a uniformly smooth, gently sloping seafloor at approximately -39 m depth: mean slope 1.0° , VRM effectively zero (10^{-5}), and no BPI anomaly. A wider 2 km scan searching for local depth anomalies (cell elevation minus 100 m neighborhood mean) found a maximum positive anomaly of only $+2.1\text{ m}$, located 900 m from the reported coordinates—far below the expected $+22\text{ m}$ signature of the pyramid.

7.2 Interpretation

The 22 m -tall Biotop pyramid is absent from the ICGC bathymetry despite being well within the survey’s spatial and depth coverage. The most likely explanation is **temporal**: the dataset version label “v2r1-2021-2025” refers to the product release, not the acquisition date for each swath. The Torredembarra area was almost certainly surveyed before the reef’s 2023 completion.

Terrain Metric Distributions — Catalan Coast Dive Sites

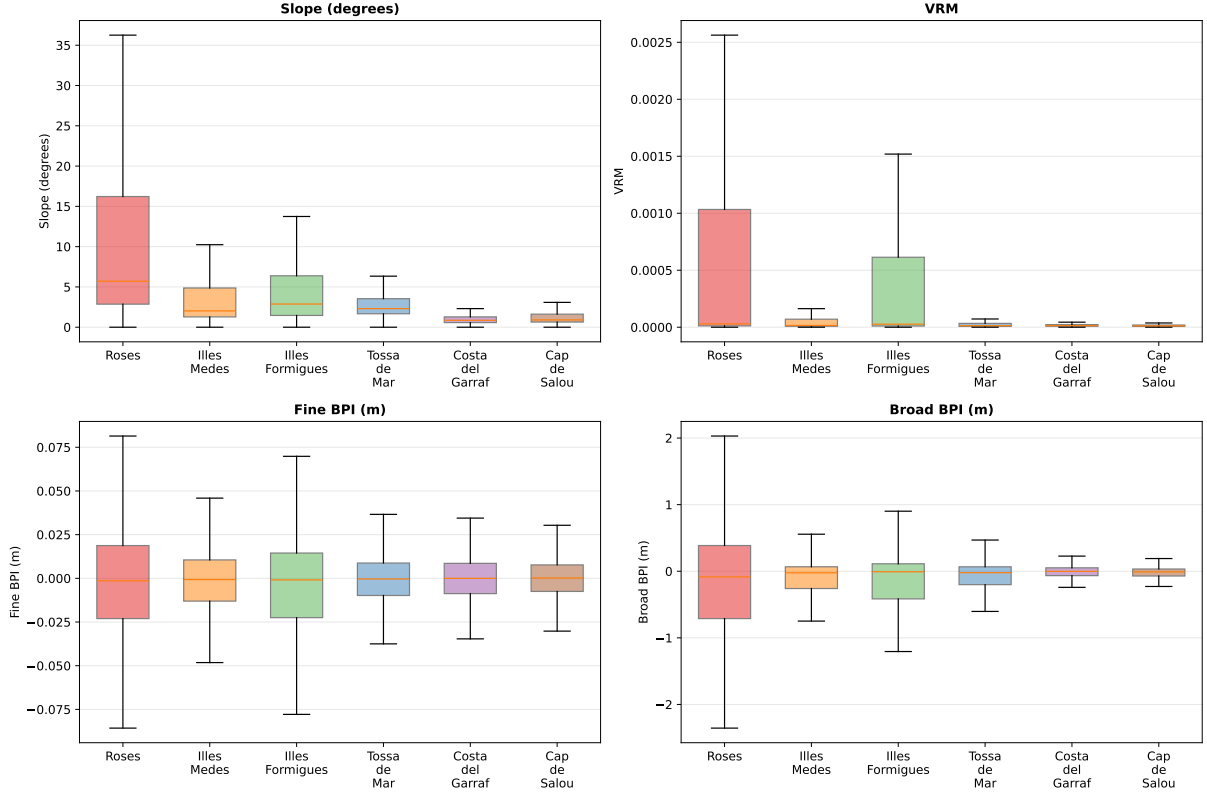


Figure 6: Box plot comparison of terrain metric distributions across six Catalan coast dive sites. Outliers suppressed for clarity. Roses consistently shows the widest distributions (steepest, most variable terrain), with Tossa de Mar and Illes Medes following.

This finding reveals an important limitation for SotaMar: the ICGC bathymetry is a **static snapshot** whose acquisition dates vary by location along the coast. Artificial reefs, recent coastal works, and other anthropogenic changes will not appear unless the area has been resurveyed after installation. Two secondary observations are also noteworthy:

- The diagonal striping visible in the fine BPI panel (Figure 7, bottom center) is a characteristic artifact of hull-mounted multibeam echosounder (MBES) surveys, caused by systematic depth offsets between adjacent swath lines. These artifacts are typically sub-meter in amplitude but become visible in high-sensitivity derivative metrics like BPI.
- The seafloor depth at the reported location (-39 m) is broadly consistent with the Biotop’s documented base depth of -34 m, suggesting the coordinates are approximately correct but that the pre-construction seabed was slightly deeper than the design specification, or the coordinates refer to a nearby point.

8 Discussion and Conclusions

This exploration validates the ICGC bathymetric dataset as an excellent foundation for the SotaMar dive site characterization system, both at the individual site level (Illes Medes pilot) and across the full Catalan coast, while also identifying critical temporal limitations through the Biotop case study. Key findings:

1. **Data quality and access.** The 1 m resolution captures dive-relevant features (walls, pinnacles, channels) at sub-meter vertical precision. The COG format enables efficient

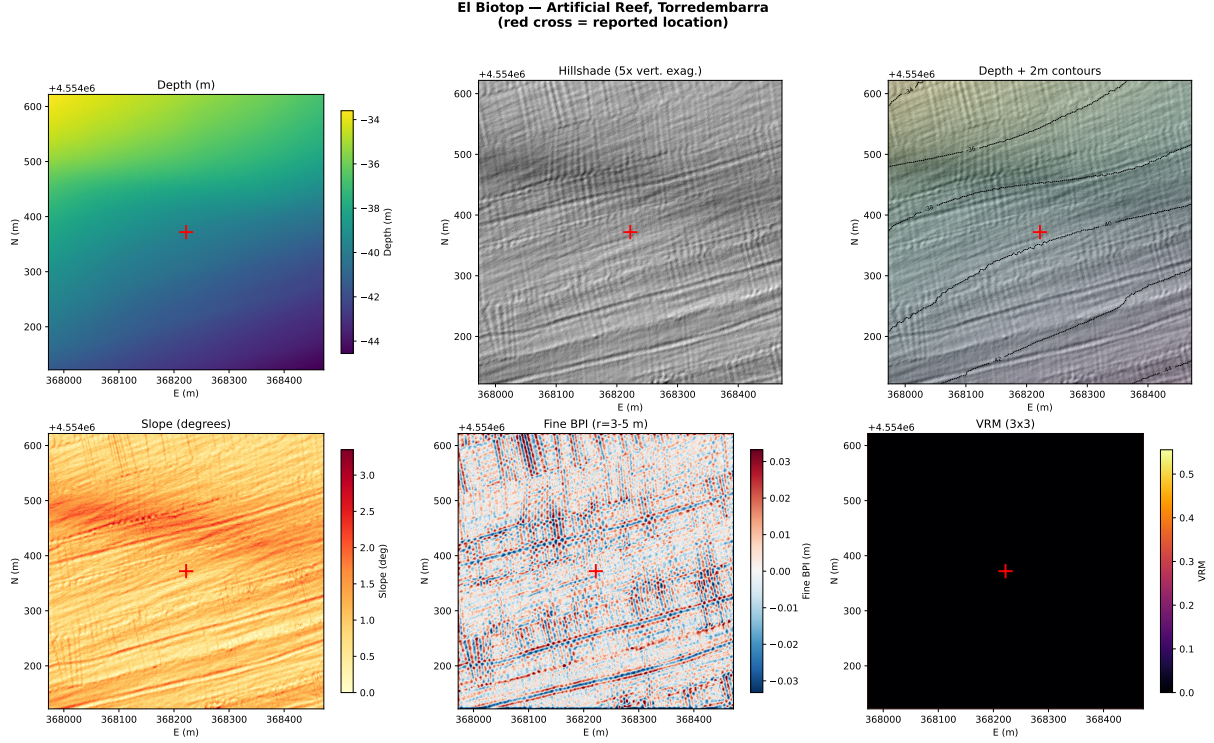


Figure 7: Terrain analysis at the reported location of El Biotop artificial reef, Torredembarra. **Top row:** depth, hillshade (5× vertical exaggeration), depth with 2m contours. **Bottom row:** slope, fine BPI, VRM. The red cross marks the reported position. No structure is visible; the seafloor is uniformly smooth. Diagonal striping in the BPI panel reveals multibeam sonar survey-line artifacts.

remote subsetting without downloading the 3.3GB file, making it practical to analyze arbitrary coastal windows on demand. Six 2km × 2km windows were extracted and analyzed in under four minutes of computation time.

2. **Metric complementarity.** No single metric fully characterizes dive site terrain. Slope identifies steep walls. Broad-scale BPI distinguishes ridges from channels. VRM isolates terrain heterogeneity. The coast-wide comparison demonstrates that these metrics are complementary: Roses leads on slope, Medes shows the highest BPI variability despite moderate slope, and Tossa de Mar ranks second on VRM despite moderate mean slope.
3. **VRM as a terrain complexity indicator.** Across all six sites, VRM provides a consistent gradient of terrain complexity, from the steepest sites (Roses: 0.005) through moderate complexity (Tossa: 0.004, island sites: 0.003) to the gentlest terrain (Garraf: 0.002). This gradient aligns well with known dive site character and could inform automated dive site scoring.
4. **Scale sensitivity.** Fine-scale BPI ($r = 3\text{--}5\text{ m}$) shows minimal variation at all sites, confirming that dive-relevant terrain structure operates at scales of 25–50 m. Terrain classification should prioritize the broad-scale BPI.
5. **Temporal limitations.** The Biotop case study demonstrates that the ICGC bathymetry is a static snapshot with spatially variable acquisition dates. Artificial reefs and recent coastal modifications will not appear unless the area has been resurveyed. SotaMar must therefore treat the ICGC data as a *baseline terrain layer* and maintain a separate registry

of known artificial structures and recent changes, cross-referenced by installation date and survey vintage.

6. **Coverage limitations.** The ICGC survey covers only the littoral fringe to approximately –60 m depth. Coverage varies from 52% (Roses) to 85% (Formigues), with all sites containing a mixture of submerged bathymetry and emerged coastal topography. Any automated pipeline must handle variable NoData density across the coast.

The next step is to implement a terrain classification scheme based on combined BPI and slope thresholds following the Benthic Terrain Modeler methodology [2], and to systematically score the full coastline for dive site potential using the VRM-based complexity gradient identified here. The Biotop case also motivates integrating temporal metadata (survey acquisition dates per swath) into the SotaMar pipeline to flag areas where the bathymetry may be outdated.

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